OpenMM Installation Guide (Ubuntu and macOS)

CIS-1 tutorial prep · Thursday, 12 Nov 2026, 2:00–3:00 PM, A3 301

Please install OpenMM on your laptop **before** the tutorial and confirm it runs using the check at the end. The recommended route is conda (Miniforge), which installs OpenMM and its dependencies in one step.

Part A: Ubuntu (20.04 / 22.04 / 24.04, x86_64)

Step 1: Install Miniforge (skip if you already have conda)

```
cd ~/Downloads
wget https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-x86_64.sh
bash Miniforge3-Linux-x86_64.sh
```

Accept the defaults, answer **yes** when asked to initialize conda, then close and reopen your terminal. You should see `(base)` at the start of the prompt.

Step 2: Create an environment and install OpenMM

```
conda create -n openmm python=3.11 -y
conda activate openmm
conda install -c conda-forge openmm -y
```

Step 3: Verify

```
python -m openmm.testInstallation
```

It prints the available platforms and a force comparison. The CPU platform is all you need. It should finish without errors.

Part B: macOS (Apple Silicon or Intel)

Step 1: Install Miniforge (skip if you already have conda)

Check your chip: Apple menu → **About This Mac**. Then download the matching installer from https://github.com/conda-forge/miniforge/releases/latest (file named `Miniforge3-MacOSX-arm64.sh` for Apple chips, `Miniforge3-MacOSX-x86_64.sh` for Intel) and run it in Terminal:

```
cd ~/Downloads
bash Miniforge3-MacOSX-arm64.sh      # use the x86_64 file on Intel Macs
```

Accept the defaults, answer **yes** to initialize conda, then close and reopen Terminal.

Step 2: Create an environment and install OpenMM

```
conda create -n openmm python=3.11 -y
conda activate openmm
conda install -c conda-forge openmm -y
```

Step 3: Verify

```
python -m openmm.testInstallation
```

It should finish without errors.

Alternative: pip (if you do not want conda)

Use a virtual environment so you do not disturb your system Python. Requires Python 3.9 or newer.

```
python3 -m venv ~/openmm-env
source ~/openmm-env/bin/activate
pip install --upgrade pip
pip install openmm
python -m openmm.testInstallation
```

On Ubuntu you may first need: `sudo apt install python3-venv python3-pip`.

Common problems

- `conda: command not found`: close and reopen your terminal. If it persists, run `~/miniforge3/bin/conda init` and reopen the terminal.
- `No module named openmm`: your environment is not active. Run `conda activate openmm` (or `source ~/openmm-env/bin/activate` for pip) and try again.
- **Installing is very slow or hangs**: make sure you are connected to a stable network; installation needs to download a few hundred MB.
- **Apple Silicon**: use the `arm64` Miniforge installer. Do not use the Intel one, it will be slower.
- **Using WSL or a VM**: follow the Ubuntu steps inside that environment.

Final check (before the tutorial)

- `python -m openmm.testInstallation` finishes without errors inside your environment
- You remember how to activate it: `conda activate openmm`
- Laptop is charged and you have brought the charger

Stuck? Message me on Moodle before the tutorial, or arrive a few minutes early to A3 301 and we will fix it together.